

TrES-2b

R-2527

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_110817 · created 2026-08-25T22:51:27+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455790.830793 +/- 9.9e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.026
Transit depth (Rp/Rs)^2	2.22 +/- 0.77 [%]
Orbital Inclination (inc)	84.73 +/- 1.49
Airmass coefficient 1 (a1)	0.96 +/- 0.039
Airmass coefficient 2 (a2)	0.035 +/- 0.049
Scatter in the residuals of the lightcurve fit is	4.0 %
Best Comparison Star	#1 - [480, 37]
Optimal Aperture	3.53
Optimal Annulus	5.01
Transit Duration (day)	0.052 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.149 $\pm$ 0.026 vs 0.1254 (deviation 0.33 σ)
Duration fit vs published	0.052 d vs 0.0746 d (deviation 0.74 σ)
Mid-time O-C	0.2 min
Depth SNR	2.1
Residual scatter	4.0 %

Light curve

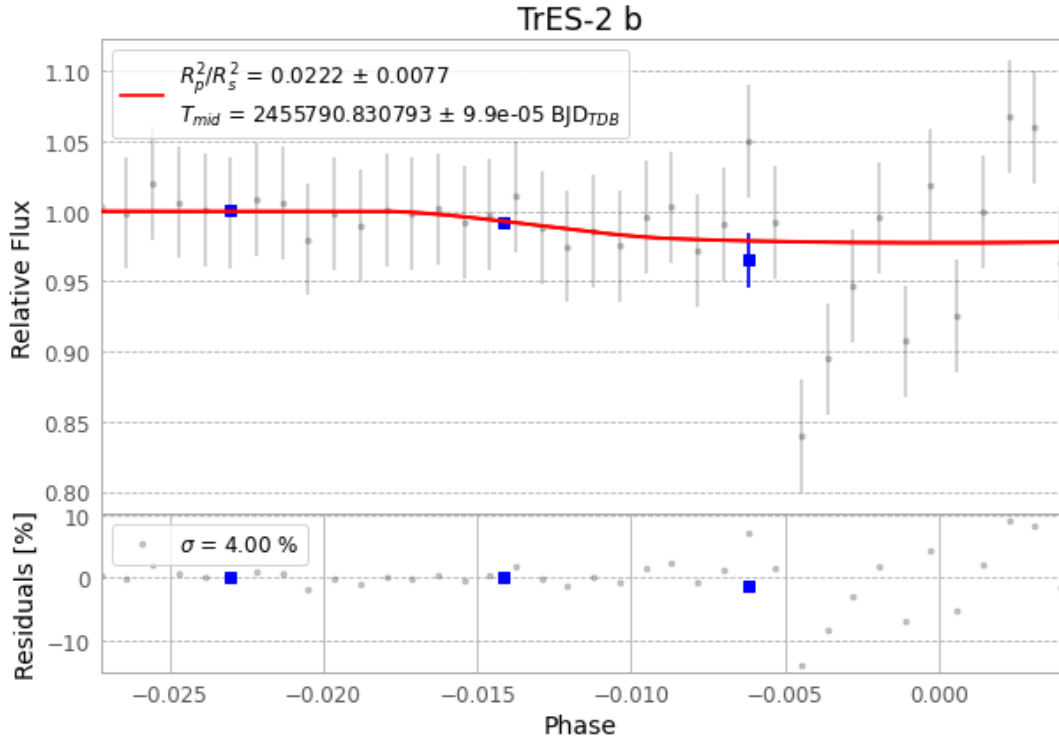


Figure 1 — FinalLightCurve_TrES-2 b_2011-08-16.png (SHA-256 4a0e84a4db644848...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [439, 310], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 92483f3db00c62e0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

70 FITS frames (46 MB), TRES-2110817061512.FITS → TRES-2110817094212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-110817.log (544ee8515705...), FinalLightCurve_TrES-2 b_2011-08-16.png (4a0e84a4db64...), FinalLightCurve_TrES-2 b_2011-08-16.csv (b0452007c2ea...)

Compute resources

Wall time 293.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 22:51Z.